Julius Kühn-Institut
Federal Research Centre for Cultivated Plants

Standardised multidimensional geodata as input data for digital soil mapping

Markus Möller

Julius Kühn-Institut · Digitalisierung und Künstliche Intelligenz · AG Forschungsdatenmanagement · Kleinmachnow

Soil Mapping

DBG annual conference on 8 September 2023

Soil Mapping

Bodenschätzung in Deutschland 10
Rückblick – Würdigung – Ausblick

Von der Bodenschätzung höchstbewertete landwirtschaftlich nutzbare Böden

Musterstück M82 mit dem Klassenzeichen L 1 Lö 100/104 (Löss-Schwarzerde); auf einer Ackerfläche des Spitzenbetriebes E. Haberhauffe, Eickendorf.

Sie befinden sich vor allem auf den Lössflächen der Magdeburger und Hildesheimer Börde sowie in der Kölner Bucht und sind mit L 1 Lö klassifiziert, z. B.

Eickendorf (FA Staßfurt)	Wertzahlen 100/104
Brumby (FA Staßfurt)	Wertzahlen 100/102
Börßum (FA Wolfenbüttel)	Wertzahlen 100/104
Wülknitz (FA Bitterfeld-Wolfen)	Wertzahlen 100/104
Nauendorf (FA Halle (S.)-Nord)	Wertzahlen 100/104
Leutenthal (FA Jena)	Wertzahlen 100/98
Rannstedt (FA Jena)	Wertzahlen 100/100
Hönnersum (FA Hildesheim)	Wertzahlen 100/104

Beschreibung
des Musterstücks für Acker Nr. 82

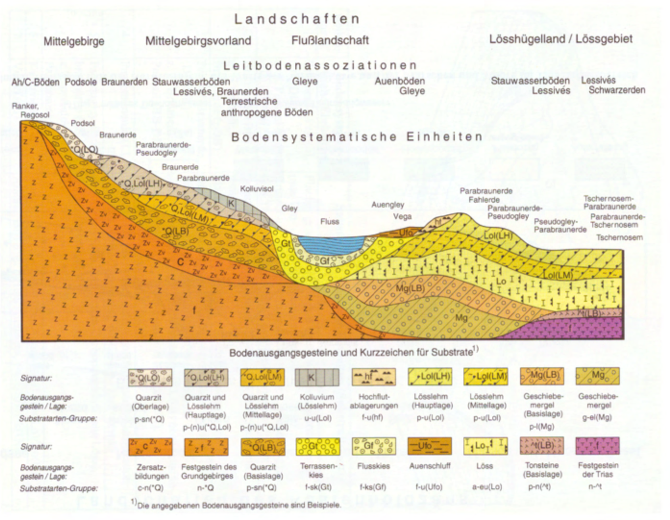
Landesfinanzamt Magdeburg — Regierungsbezirk Magdeburg
Finanzamt Schönebeck a/Elbe — Kreis Calbe
Katasteramt Schönebeck a/Elbe — Gemeinde Eickendorf

Tag der Besichtigung	Gemarkung	Planblatt	Parzelle	Eigentümer	Größe ha	Größe a	Klima Seehöhe m	Klima Jahresniederschläge mm	Klima Jahreswärme °C	Bodengefüge
1	2	3	4	5	6		7	8	9	10
18.5 1934	Eickendorf	2	547/54	Haberhauffe, Witwe Else geb. Wiegand	.	62	87 / 84	483 / 444	9	h mi L3 / h mi L2 / ka - L6

Klasse Bodenart	Klasse Zustandsstufe	Klasse Entstehung	Bodenzahl	Besonderheiten (Verschieden, Geländegestaltung u.a.)	Besonderheiten z.u.H.	Allgemeines Klima z.u.H.	Ackerzahl	Bemerkungen
11	12	13	14	15	16	17	18	
L	1	Lö	100			+4	106 / 104	

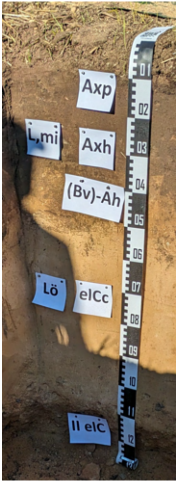
Soil Mapping

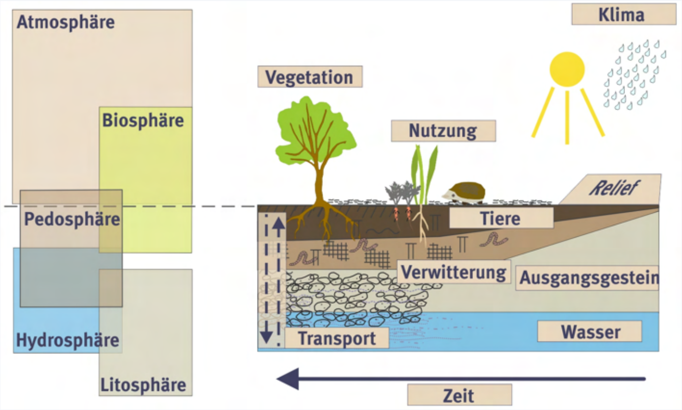
Dimensions



Ad-hoc-Arbeitsgruppe Boden, 2005. Bodenkundliche Kartieranleitung, 5.Auflage. Schweizerbart'sche Verlagsbuchhandlung (Nägele und Obermiller), Stuttgart.

Soil Mapping





UBA, 2010. Die Böden Deutschlands – Sehen, Erkunden, Verstehen. Umweltbundesamt.

Digital Soil Mapping

The CLORPT model

$$s = f(cl, o, r, p, t, \dots) \tag{1}$$

with s = soil, cl = climate, o = organisms, r = relief, p = parent material, t = time, $(\dots)$ = unique factors

The SCORPAN model

$$S = f(s, c, o, r, p, a, n) + e \tag{2}$$

with S = soil attributes or classes, c = climate, o = organisms (including land cover and natural vegetation), r = topography, p = parent material, a = age or time, n = spatial location or position, e = spatially correlated residuals

CLORPT model was designed to produce knowledge, whereas SCORPAN was not intended for explaining soil formation, but rather a pragmatic empirical model for predicting soil properties and soil classes (production of maps).

Ma, Y., Minasny, B., Malone, B.P., Mcbratney, A.B., 2019. Pedology and digital soil mapping (DSM). European J Soil Science 70, 216–235. https://doi.org/10.1111/ejss.12790

Digital Soil Mapping

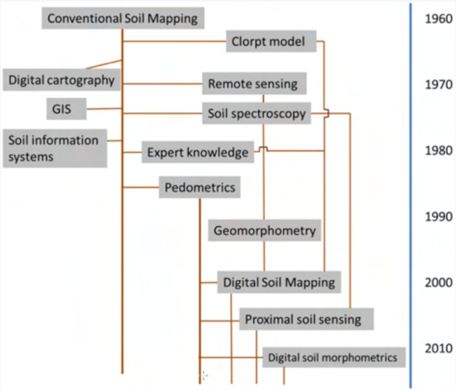
Paradigms

	Expert/knowledge-driven soil mapping	Data/technology-driven soil mapping
Target variables:	Soil types (soil series)	Analytical soil properties
Spatial data model:	Discrete (soil bodies)	Continuous/hybrid (quantities / probabilities)
Major inputs:	Expert knowledge / soil profile description	Laboratory data / proximal soil sensing
Important covariates:	Soil delineations (photo-interpretation)	Remote sensing images, DEM-derivatives
Spatial prediction model:	Averaging per polygon	Automated (geo)statistics
Accuracy assessment:	Validation of soil mapping units (kappa)	Cross-validation (RMSE)
Data representation:	Polygon maps + attribute tables	Gridded maps + prediction error map
Major technical aspect:	Cartographic scale	Grid cell size
Soil sampling strategies:	Free survey (surveyor selects sampling)	Statistical (design/model-based) sampling)

Hengl, T., MacMillan, R.A., 2019. Predictive Soil Mapping with R. OpenGeoHub foundation, Wageningen. `https://www.soilmapper.org`, ISBN: 978-0-359-30635-0

Digital Soil Mapping

Concepts



Minasny, B., McBratney, Alex.B., 2016. Digital soil mapping: A brief history and some lessons. Geoderma 264, 301–311. `https://doi.org/10.1016/j.geoderma.2015.07.017`

Definition

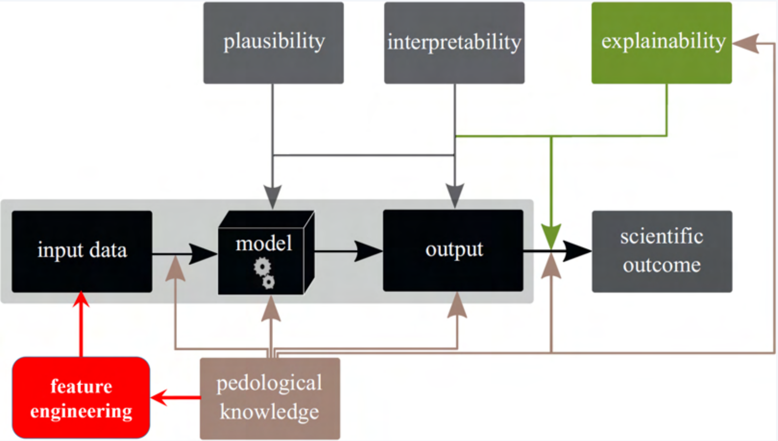
... the creation and population of spatial soil information systems by the use of field and laboratory observational methods coupled with spatial and non-spatial soil inference systems

Digital soil morphometrics

... aims to yield new insights in soil horizonation and enhance pedological understanding.

Digital Soil Mapping

Explainable Machine Learning



Wadoux, A.M.J.-C., Minasny, B., McBratney, A.B., 2020. Machine learning for digital soil mapping: Applications, challenges and suggested solutions. Earth-Science Reviews 210, 103359.
https://doi.org/10.1016/j.earscirev.2020.103359

Digital Soil Mapping

Explainable Machine Learning

Transparency considers the ML approach, interpretability considers the ML model together with data, and explainability considers the model, the data, and human involvement.

Feature engineering

- adjusting and reworking the predictors to enable models to better uncover predictor-response relationships
- the process of creating representations of data that increase the effectiveness of a model

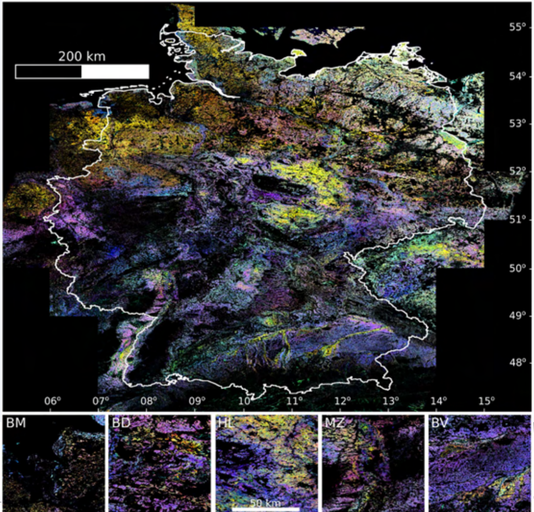
Roscher, R., Bohn, B., Duarte, M.F., Garcke, J., 2020. Explainable Machine Learning for Scientific Insights and Discoveries. IEEE Access 8, 42200–42216. https://doi.org/10.1109/ACCESS.2020.2976199

Kuhn, M., Johnson, K., 2019. Feature Engineering and Selection: A Practical Approach for Predictive Models, 1st ed. Chapman and Hall/CRC. https://doi.org/10.1201/9781315108230

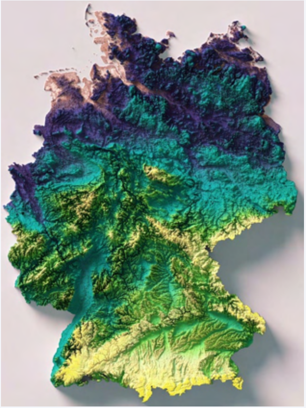
Feature engineering

Bare Soil Composites

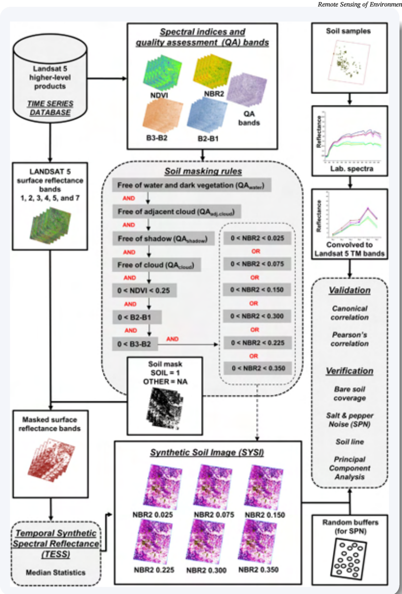
Rogge et al. Remote Sensing of Environment 205 (2018



Terrain attributes



Feature engineering: Time



g. 4. Flowchart of the Geospatial Soil Sensing System (GEOS3) and its products: Temporal Synthetic Spectral Reflectance (TESS) and Synthetic Soil Image (SYSI).
DVI: Normalized Difference Vegetation Index. NBR2: Normalized Burn Ratio 2. The terms AND and OR are Boolean operators used in the script of GEOS3. The AND

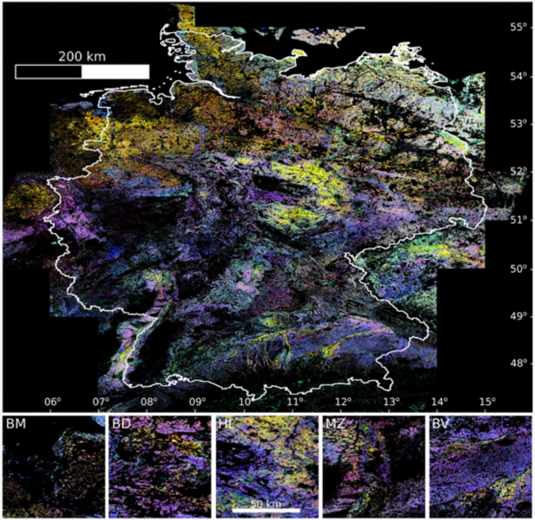
Flowchart for the derivation of Bare Soil Composites (BSC)

- TESS – Temporal Synthetic Spectral Reflectance
- SYSI – Synthetic Soil Image
- NDVI – Normalized Difference Vegetation Index
- NBR2 – Normalized Burn Ratio

Demattê, J.A.M., Fongaro, C.T., Rizzo, R., Safanelli, J.L., 2018. Geospatial Soil Sensing System (GEOS3): A powerful data mining procedure to retrieve soil spectral reflectance from satellite images. Remote Sensing of Environment 212, 161–175. https://doi.org/10.1016/j.rse.2018.04.047

Feature engineering: Time

D. Rogge et al. Remote Sensing of Environment 205 (2018) 1–17



BSC (bands 7-5-3) for Germany and for five investigation areas derived from 2000 to 2004 Landsat imagery

Rogge, D., Bauer, A., Zeidler, J., Mueller, A., Esch, T., Heiden, U., 2018. Building an exposed soil composite processor (SCMaP) for mapping spatial and temporal characteristics of soils with Landsat imagery (1984–2014). Remote Sensing of Environment 205, 1–17. https://doi.org/10.1016/j.rse.2017.11.004

Fig. 10. *SRCnorm* soil composite image (Landsat bands 7-5-3) for the study area and for the five investigation areas derived from 2000 to 2004 Landsat imagery (Geographic Lat-Long projection, WGS84). Outline of Germany shown in white. Scale bar calculated based on 51.5° north.

minimize mixing with other classes, have multiple sample polygons for with limited exposed soil. For SCMaP, the extraction of exposed soils

Feature engineering: Time

D. Rogge et al. Remote Sensing of Environment 205 (2018) 1–17

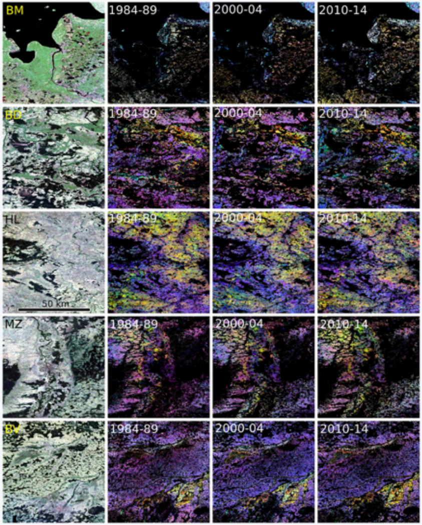


Fig. 11. *SRCnorm* soil composite image (Landsat bands 7-5-3) for the five investigation areas over three time periods showing subtle variations in localized regions that may be linked to changing soil parameters. Included in the far left column are the true color RGB images (Landsat bands 3-2-1) for reference (Geographic Lat-Long projection, WGS84). Scale bar calculated based on 51.5° north. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

BSCs (Landsat bands 7-5-3) for five investigation areas over three time periods showing subtle variations in localized regions that may be linked to changing soil parameters. Included in the far left column are the true color RGB images (Landsat bands 3-2-1) for reference.

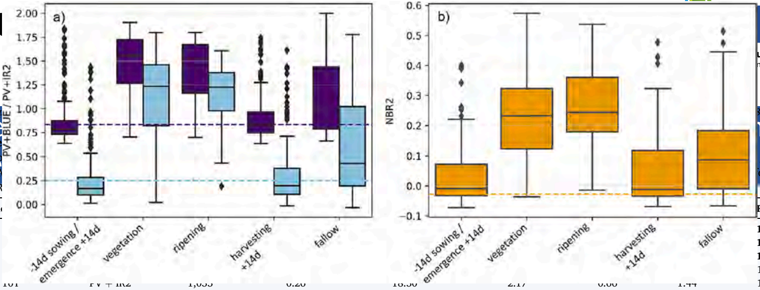
Rogge, D., Bauer, A., Zeidler, J., Mueller, A., Esch, T., Heiden, U., 2018. Building an exposed soil composite processor (SCMaP) for mapping spatial and temporal characteristics of soils with Landsat imagery (1984–2014). Remote Sensing of Environment 205, 1–17. `https://doi.org/10.1016/j.rse.2017.11.004`

version of this article.)

Feat…

Pe… me

Table …
Numb… given … R², RM… for the best R…

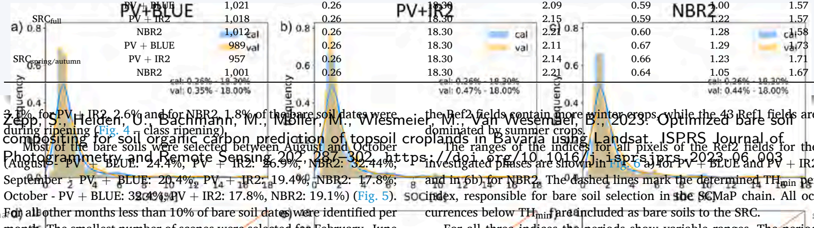


Tim…								RPD
								1.91
								1.87
								1.76
								1.79
	PV + IR2		0.26	18.30	2.17	0.66	1.44	1.73
	NBR2	1,008	0.26	18.30	2.18	0.61	1.34	1.61
	PV + BLUE	995	0.26	18.30	2.14	0.65	1.09	1.68
7Y	PV + IR2	981	0.26	18.30	2.16	0.61	1.42	1.60
	NBR2	1,005	0.26	18.30	2.18	0.61	1.21	1.60
	PV + BLUE	989	0.26	18.30	2.11	0.67	1.29	1.73
5Y	PV + IR2	957	0.26	18.30	2.14	0.66	1.23	1.71
	NBR2	1,001	0.26	18.30	2.21	0.64	1.05	1.67
	PV + BLUE	932	0.26	18.30	2.09	0.56	1.19	1.51
3Y	PV + IR2	832	0.26	18.30	2.05	0.61	1.29	1.60
	NBR2	884	0.26	18.30	2.14	0.66	1.33	1.70

Fig. 6. Index ranges for all cloudless scenes within the different summarized phenological phases. Shown are the 30 sample fields of Ref2 dataset. The dashed lines indicate the derived TH_{min} used for bare soil definition of the respective index. All occurrences below TH_{min} are included to the SRC generation (sowing, veg = period with photosynthetically active vegetation cover, ripening, harvesting, fallow) of Ref2 for a) PV + BLUE (indigo), PV + IR2 (light blue) and b) NBR2 (orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Intersecting samples for modeling, SOC statistics and model performances based on SRC_{full} and $SRC_{spring/autumn}$ for the compositing period of 2015–2019 for all three tested indices. Shown are the R^2, RMSE and RPD values of model validation (30% of input data).

2015–19	index	samples	SOC min [%]	SOC max [%]	SOC mean [%]	R^2	RMSE [%]	RPD
	PV + BLUE	1,021	0.26	18.30	2.09	0.59	1.00	1.57
SRC_{full}	PV + IR2	1,018	0.26	18.30	2.15	0.59	1.22	1.57
	NBR2	1,012	0.26	18.30	2.21	0.60	1.28	1.58
	PV + BLUE	989	0.26	18.30	2.11	0.67	1.29	1.73
$SRC_{spring/autumn}$	PV + IR2	957	0.26	18.30	2.14	0.66	1.23	1.71
	NBR2	1,001	0.26	18.30	2.21	0.64	1.05	1.67



3.1% for PV + IR2, 2.6% and for NBR2, 1.8% of the bare soil dates were during ripening (Fig. 4 – class ripening).

Most of the bare soils were selected between August and October (August - PV + BLUE: 24.4%, PV + IR2: 26.9%, NBR2: 32.44%; September - PV + BLUE: 20.4%, PV + IR2: 19.4%, NBR2: 17.8%; October - PV + BLUE: 32.4%, PV + IR2: 17.8%, NBR2: 19.1%) (Fig. 5). For all other months less than 10% of bare soil dates were identified per month. The smallest number of scenes were selected for February, June

the Ref2 fields contain more winter crops, while the 43 Ref1 fields are dominated by summer crops.

The ranges of the indices for all pixels of the Ref2 fields for the investigated phases are shown in Fig. 6 a) for PV + BLUE and PV + IR2 and in 6b) for NBR2. The dashed lines mark the determined TH_{min} per index, responsible for bare soil selection in the SCMaP chain. All occurrences below TH_{min} are included as bare soils to the SRC.

For all three indices the periods show variable ranges. The period

Zepp, S., Heiden, U., Bachmann, M., Möller, M., Wiesmeier, M., Van Wesemael, B., 2023. Optimized bare soil compositing for soil organic carbon prediction of topsoil croplands in Bavaria using Landsat. ISPRS Journal of Photogrammetry and Remote Sensing 202, 287–302. https://doi.org/10.1016/j.isprsjprs.2023.06.003

Feature engineering: Scale

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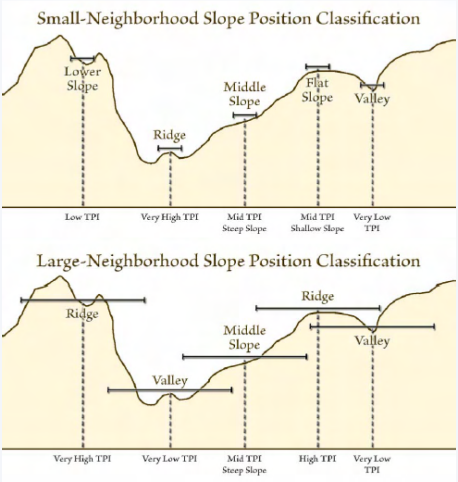
Federal Research Centre for Cultivated Plants

Multi-scale landforms



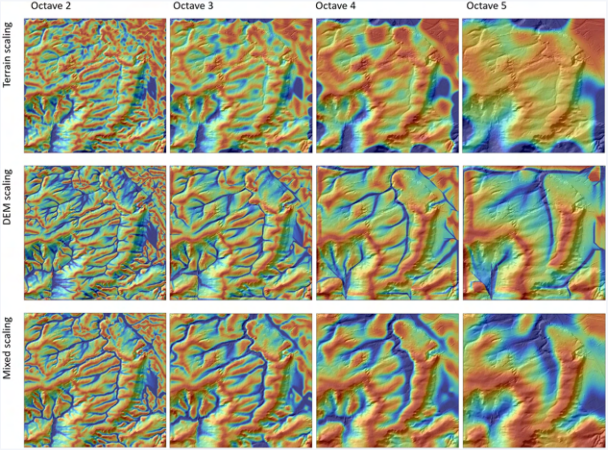
Concepts

1. DEM scaling
2. terrain scaling (moving window filter sizes)
3. segmentation

- Guisan, A., Weiss, S.B., Weiss, A.D., 1999. GLM versus CCA spatial modeling of plant species distribution. Plant Ecology 143, 107–122. https://doi.org/10.1023/A:1009841519580
- Behrens, T., Schmidt, K., Ramirez-Lopez, L., Gallant, J., Zhu, A.-X., Scholten, T., 2014. Hyper-scale digital soil mapping and soil formation analysis. Geoderma 213, 578–588. https://doi.org/10.1016/j.geoderma.2013.07.031
- Minár, J., Drăguț, L., Evans, I.S., Feciskanin, R., Gallay, M., Jenčo, M., Popov, A., 2024. Physical geomorphometry for elementary land surface segmentation and digital geomorphological mapping. Earth-Science Reviews 248, 104631. https://doi.org/10.1016/j.earscirev.2023.104631

Feature engineering: Scale

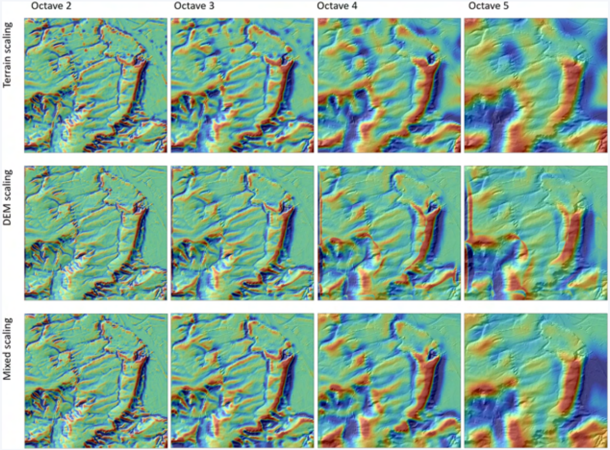
Flow accumulation



Behrens, T., Viscarra Rossel, R.A., Kerry, R., MacMillan, R., Schmidt, K., Lee, J., Scholten, T., Zhu, A.-X., 2019. The relevant range of scales for multi-scale contextual spatial modelling. Sci Rep 9, 14800. `https://doi.org/10.1038/s41598-019-51395-3`

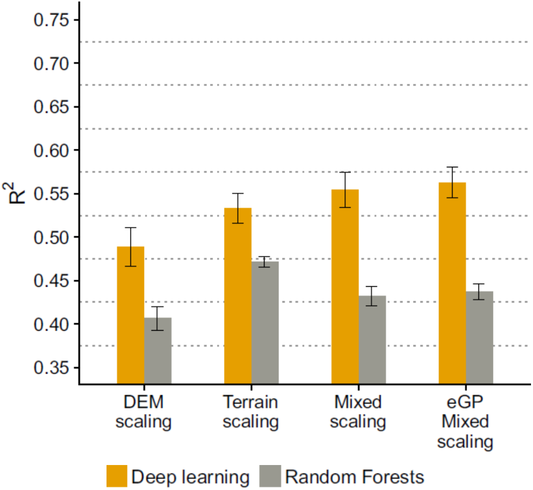
Feature engineering: Scale

Cross curvarure



Behrens, T., Viscarra Rossel, R.A., Kerry, R., MacMillan, R., Schmidt, K., Lee, J., Scholten, T., Zhu, A.-X., 2019. The relevant range of scales for multi-scale contextual spatial modelling. Sci Rep 9, 14800. https://doi.org/10.1038/s41598-019-51395-3

Feature engineering: Scale

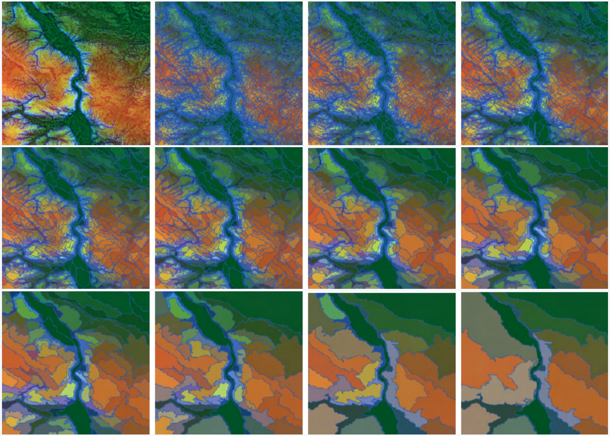


Modelling accuracy on the example of log-transformed zinc concentration

Behrens, T., Viscarra Rossel, R.A., Kerry, R., MacMillan, R., Schmidt, K., Lee, J., Scholten, T., Zhu, A.-X., 2019. The relevant range of scales for multi-scale contextual spatial modelling. Sci Rep 9, 14800. `https://doi.org/10.1038/s41598-019-51395-3`

Feature engineering: Scale

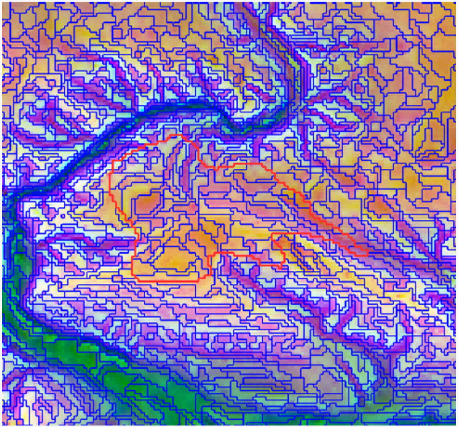
Scale levels based on the segmentation of terrain attributes

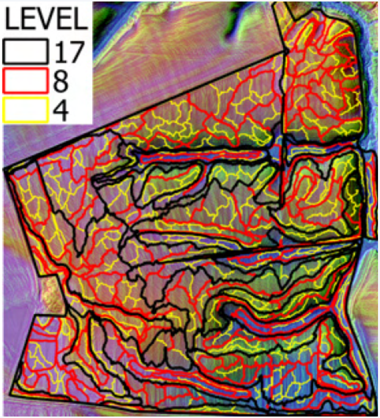


- Elevation
- Vertical distance to channel net
- Slope
- Cross curvature

Möller, M., Volk, M., Friedrich, K., Lymburner, L., 2008. Placing soil-genesis and transport processes into a landscape context: A multiscale terrain-analysis approach. J. Plant Nutr. Soil Sci. 171, 419–430. https://doi.org/10.1002/jpln.200625039

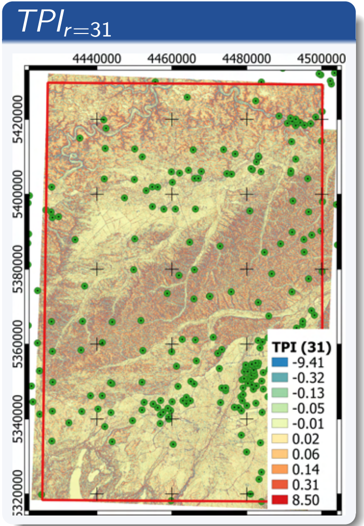
Feature engineering: Scale

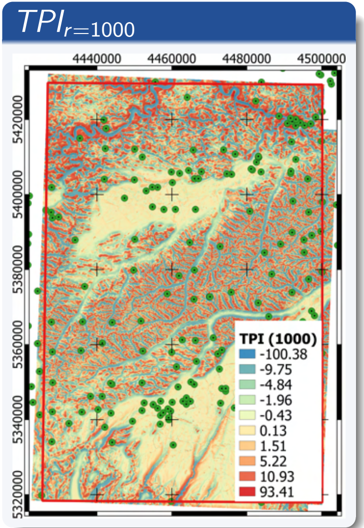




- Möller M, Koschitzki T, Hartmann K-J, Jahn R. Plausibility test of conceptual soil maps using relief parameters. CATENA 2012;88:57–67. https://doi.org/10.1016/j.catena.2011.08.002.
- Möller M, Volk M. Effective map scales for soil transport processes and related process domains — Statistical and spatial characterization of their scale-specific inaccuracies. Geoderma 2015;247–248:151–60. https://doi.org/10.1016/j.geoderma.2015.02.003 .

Feature engineering: Scale

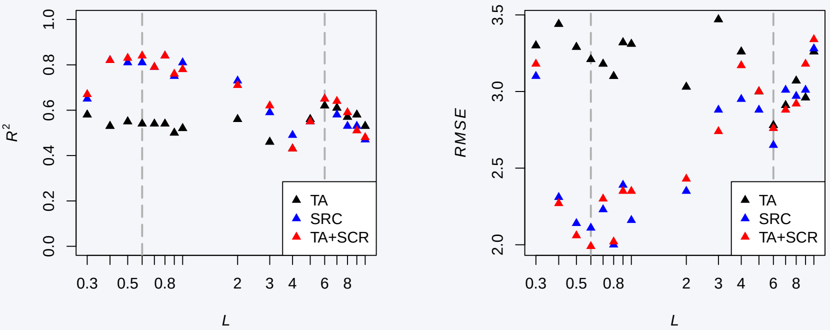




Möller, M., Zepp, S., Wiesmeier, M., Gerighausen, H., Heiden, U., 2022. Scale-Specific Prediction of Topsoil Organic Carbon Contents Using Terrain Attributes and SCMaP Soil Reflectance Composites. Remote Sensing 14, 2295. `https://doi.org/10.3390/rs14102295`

Feature engineering: Scale

SOC prediction



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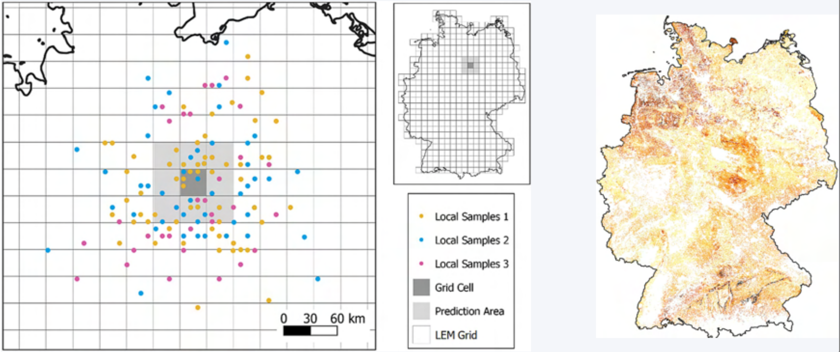
Feature engineering: Space

Local Ensemble Modelling



Broeg, T., Don, A., Gocht, A., Scholten, T., Taghizadeh-Mehrjardi, R., Erasmi, S., 2024. Using local ensemble models and Landsat bare soil composites for large-scale soil organic carbon maps in cropland. Geoderma 444, 116850. https://doi.org/10.1016/j.geoderma.2024.116850

Feature engineering: Space

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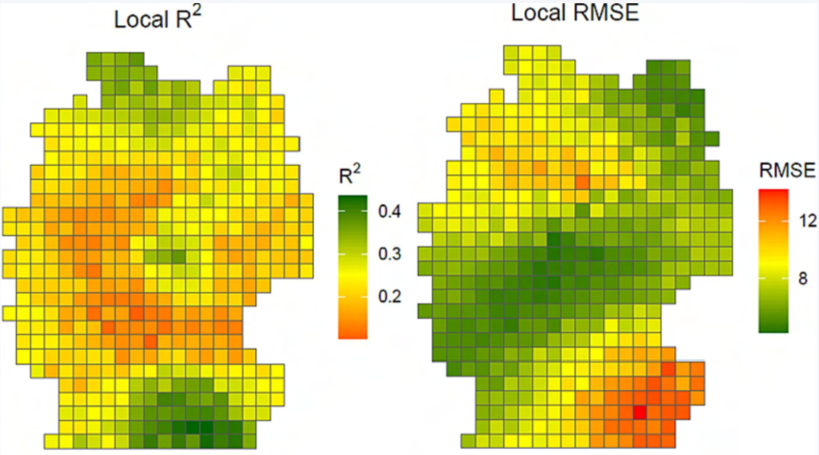
Federal Research Centre for Cultivated Plants

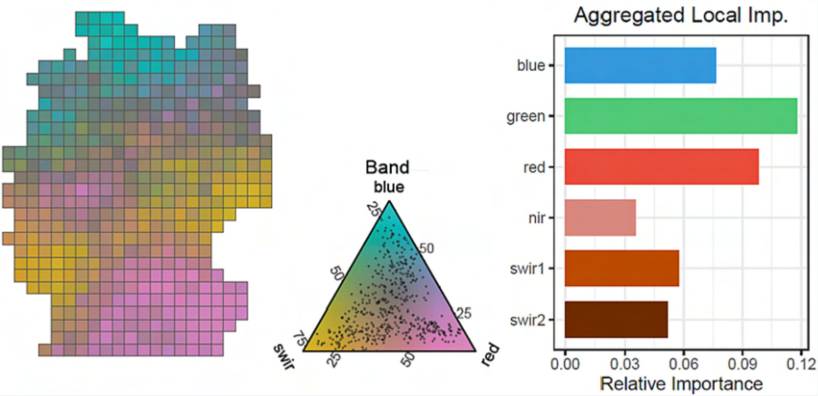
Local Ensemble Modelling



Broeg, T., Don, A., Gocht, A., Scholten, T., Taghizadeh-Mehrjardi, R., Erasmi, S., 2024. Using local ensemble models and Landsat bare soil composites for large-scale soil organic carbon maps in cropland. Geoderma 444, 116850. https://doi.org/10.1016/j.geoderma.2024.116850

Feature engineering: Space

Local Ensemble Modelling



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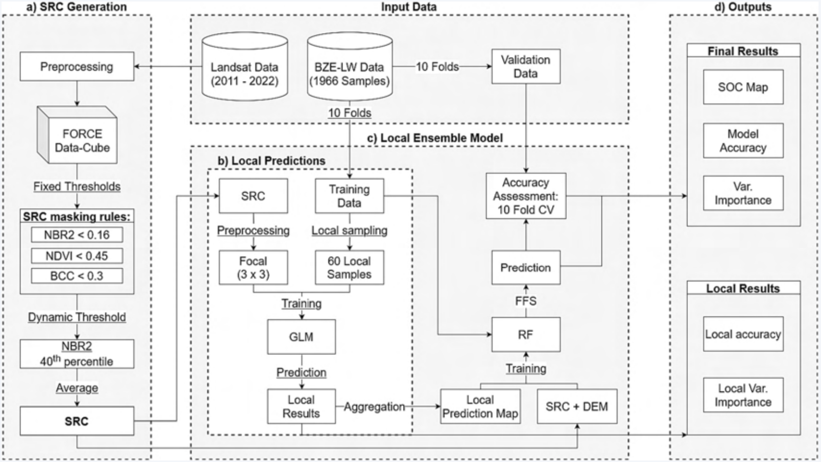
Feature engineering: Space

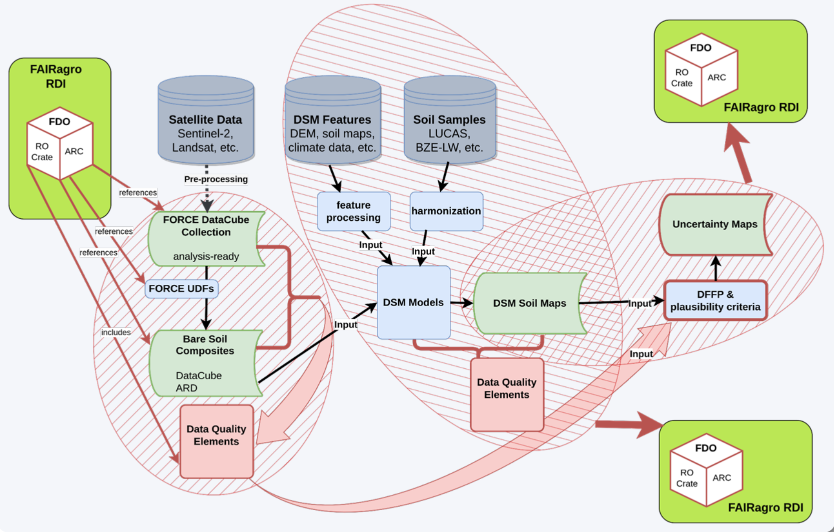
Local Ensemble Modelling



Broeg, T., Don, A., Gocht, A., Scholten, T., Taghizadeh-Mehrjardi, R., Erasmi, S., 2024. Using local ensemble models and Landsat bare soil composites for large-scale soil organic carbon maps in cropland. Geoderma 444, 116850. https://doi.org/10.1016/j.geoderma.2024.116850

Feature engineering: Space

FAIR lineage tracking



Conclusion

Julius Kühn-Institut
Federal Research Centre for Cultivated Plants

Feature engineering . . .

- represents dynamic variable characteristics in relation to space and time
- can formalize expert and domain knowledge and perceptions,
- can improve model quality and acceptance.

Conclusion

Julius Kühn-Institut
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Feature engineering ...

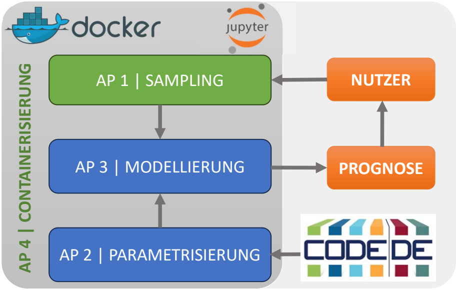
- represents dynamic variable characteristics in relation to space and time
- can formalize expert and domain knowledge and perceptions,
- can improve model quality and acceptance.

Towards Germany-wide FAIR explaining variables!

- Germany-wide basic data and technical big data requirements for state and federal authorities are in place (⇒ CODE-DE).
- System for managing research data is under development (⇒ FAIRagro and can be actively influenced!)

Conclusion

Dynamic prediction model of soil



(Geo)Data Life Cycle

Questions?